

Elementary Number Theory

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Contents

Preface	v
1 Preliminaries	1
1.1 Mathematical Induction	1
1.2 The Binomial Theorem	1
2 Divisibility Theory in the Integers	2
2.1 Early Number Theory	2
2.2 The Division Algorithm	2
2.3 The Greatest Common Divisor	4
2.4 The Euclidean Algorithm	4
2.5 The Diophantine Equation $ax+by=c$	4
3 Primes and Their Distribution	5
3.1 The Fundamental Theorem of Arithmetic	5
3.2 The Sieve of Eratosthenes	5
3.3 The Goldbach Conjecture	5
4 The Theory of Congruences	6
4.1 Carl Friedrich Gauss	6
4.2 Basic Properties of Congruence	6
4.3 Binary and Decimal Representations of Integers	6
4.4 Linear Congruences and the Chinese Remainder Theorem	6
5 Fermat's Theorem	7
5.1 Pierre de Fermat	7
5.2 Fermat's Little Theorem and Pseudoprimes	7
5.3 Wilson's Theorem	7
5.4 The Fermat-Kraitichik Factorization Method	7
6 Number-Theoretic Functions	8
6.1 The Sum and Number of Divisors	8
6.2 The Möbius Inversion Formula	8

6.3	The Greatest Integer Function	8
6.4	An Application to the Calendar	8
7	Euler's Generalization of Fermat's Theorem	9
7.1	Leonhard Euler	9
7.2	Euler's Phi-Function	9
7.3	Euler's Theorem	9
7.4	Some Properties of the Phi-Function	9
8	Primitive Roots and Indices	10
8.1	The Order of an Integer Modulo n	10
8.2	Primitive Roots for Primes	10
8.3	Composite Numbers Having Primitive Roots	10
8.4	The Theory of Indices	10
9	The Quadratic Reciprocity Law	11
9.1	Euler's Criterion	11
9.2	The Legendre Symbol and Its Properties	11
9.3	Quadratic Reciprocity	11
9.4	Quadratic Congruences with Composite Moduli	11
10	Introduction to Cryptography	12
10.1	From Caesar Cipher to Public Key Cryptography	12
10.2	The Knapsack Cryptosystem	12
10.3	An Application of Primitive Roots to Cryptography	12
11	Numbers of Special Form	13
11.1	Marin Mersenne	13
11.2	Perfect Numbers	13
11.3	Mersenne Primes and Amicable Numbers	13
11.4	Fermat Numbers	13
12	Certain Nonlinear Diophantine Equations	14
12.1	The Equation $x^2 + y^2 = z^2$	14
12.2	Fermat's Last Theorem	14
13	Representation of Integers as Sums of Squares	15
13.1	Joseph Louis Lagrange	15
13.2	Sums of Two Squares	15
13.3	Sums of More Than Two Squares	15

14	Fibonacci Numbers	16
14.1	Fibonacci	16
14.2	The Fibonacci Sequence	16
14.3	Certain Identities Involving Fibonacci Numbers	16
15	Continued Fractions	17
15.1	Srinivasa Ramanujan	17
15.2	Finite Continued Fractions	17
15.3	Infinite Continued Fractions	17
15.4	Farey Fractions	17
15.5	Pell’s Equation	17
16	Some Modern Developments	18
16.1	Hardy, Dickson, and Erdős	18
16.2	Primality Testing and Factorization	18
16.3	An Application to Factoring: Remote Coin Flipping	18
16.4	The Prime Number Theorem and Zeta Function	18

Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

Chapter 1

Preliminaries

1.1 Mathematical Induction

1.2 The Binomial Theorem

Chapter 2

Divisibility Theory in the Integers

2.1 Early Number Theory

2.2 The Division Algorithm

Theorem 2.2.1.

Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}^+$. There is a unique pair $(q, r) \in \mathbb{Z} \times \mathbb{N}$ such that

$$a = qb + r, \quad r < b.$$

We call q the **quotient** and r the **remainder**.

```
theorem existsUnique_quotient_remainder (a : ℤ) (b : ℕ+) :  
  ∃! qr : ℤ × ℕ, a = qr.1 * b + qr.2 ∧ qr.2 < b
```

Proof.

Since $b \geq 1$,

$$a - (-|a|)b = a + |a|b \geq a + |a| \geq 0.$$

```
classical  
have hb : (1 : ℤ) ≤ (b : ℤ) := by exact_mod_cast b.pos  
have hnonneg : 0 ≤ a - (-|a|) * (b : ℤ) := by  
  have ha : -|a| ≤ a := neg_abs_le a  
  have habs : 0 ≤ |a| := abs_nonneg a  
  nlinarith
```

Define

$$S := \{n \in \mathbb{N} : a - qb = n \text{ for some } q \in \mathbb{Z}\}.$$

By the preceding inequality, $S \neq \emptyset$. Let $r = \min S$ and choose $q \in \mathbb{Z}$ with $a - qb = r$.

```

let h :  $\exists n : \mathbb{N}, \exists q : \mathbb{Z}, a - q * b = (n : \mathbb{Z})$  :=
  ((a - (-|a|) * b).toNat, -|a|, (Int.toNat_of_nonneg hnonneg).symm)
obtain ⟨q, hq⟩ := Nat.find_spec h
let r :  $\mathbb{N}$  := Nat.find h
have hr0 : (0 :  $\mathbb{Z}$ ) ≤ r := Int.natCast_nonneg _

```

By minimality, for $s \in \mathbb{Z}$,

$$0 \leq s, \quad s = a - xb \ (x \in \mathbb{Z}) \implies r \leq s.$$

```

have hminimal :  $\forall s : \mathbb{Z}, 0 \leq s \rightarrow (\exists x : \mathbb{Z}, a - x * b = s) \rightarrow (r : \mathbb{Z}) \leq s$ 
  ↪ := by
    intro s hs ⟨x, hx⟩
    have hn : Nat.find h ≤ s.toNat :=
      Nat.find_min' h ⟨x, hx.trans (Int.toNat_of_nonneg hs).symm⟩
    have hc : (Nat.find h :  $\mathbb{Z}$ ) ≤ (s.toNat :  $\mathbb{Z}$ ) := Int.ofNat_le.mpr hn
    rwa [Int.toNat_of_nonneg hs] at hc

```

If $r \geq b$, then

$$0 \leq r - b = a - (q + 1)b < r,$$

contradicting minimality. Hence $r < b$.

```

have hrb_int : (r :  $\mathbb{Z}$ ) < b := by
  by_contra hnot
  have hs : 0 ≤ (r :  $\mathbb{Z}$ ) - b := by linarith
  have hsmaller := hminimal ((r :  $\mathbb{Z}$ ) - b) hs ⟨q + 1, by dsimp [r];
    ↪ nlinarith [hq]⟩
  linarith

```

Thus $a = qb + r$ and $0 \leq r < b$, proving existence.

```

have ha : a = q * b + r := by dsimp [r]; linarith [hq]
have hrb : r < (b :  $\mathbb{N}$ ) := by exact_mod_cast hrb_int
refine ⟨(q, r), ⟨ha, hrb⟩, ?_⟩

```

Suppose another pair $(q', r') \in \mathbb{Z} \times \mathbb{N}$ satisfies $a = q'b + r'$ and $0 \leq r' < b$.

```
rintro (q', r') (ha', hr'b)
change a = q' * b + r' at ha'
have hr'0 : (0 : ℤ) ≤ r' := Int.natCast_nonneg _
have hr'b_int : (r' : ℤ) < b := by exact_mod_cast hr'b
```

Since $q, q' \in \mathbb{Z}$,

$$\begin{aligned} q' < q &\implies q - q' \geq 1 \implies r' = r + (q - q')b \geq b, \\ q' > q &\implies q' - q \geq 1 \implies r = r' + (q' - q)b \geq b. \end{aligned}$$

Both contradict the remainder bounds, so $q' = q$.

```
have hqq : q' = q := by
  rcases lt_trichotomy q' q with hlt | heq | hgt
  · have hstep : q' + 1 ≤ q := hlt
    nlinarith
  · exact heq
  · have hstep : q + 1 ≤ q' := hgt
    nlinarith
```

Then $r' = a - q'b = a - qb = r$, so $(q', r') = (q, r)$, proving uniqueness.

```
have hrr_int : (r' : ℤ) = r := by rw [hqq] at ha'; linarith
have hrr : r' = r := by exact_mod_cast hrr_int
exact Prod.ext hqq hrr
```

□

2.3 The Greatest Common Divisor

2.4 The Euclidean Algorithm

2.5 The Diophantine Equation $ax + by = c$

Chapter 3

Primes and Their Distribution

3.1 The Fundamental Theorem of Arithmetic

3.2 The Sieve of Eratosthenes

3.3 The Goldbach Conjecture

Chapter 4

The Theory of Congruences

4.1 Carl Friedrich Gauss

4.2 Basic Properties of Congruence

4.3 Binary and Decimal Representations of Integers

4.4 Linear Congruences and the Chinese Remainder Theorem

Chapter 5

Fermat's Theorem

5.1 Pierre de Fermat

5.2 Fermat's Little Theorem and Pseudoprimes

5.3 Wilson's Theorem

5.4 The Fermat-Kraitchik Factorization Method

Chapter 6

Number-Theoretic Functions

6.1 The Sum and Number of Divisors

6.2 The Möbius Inversion Formula

6.3 The Greatest Integer Function

6.4 An Application to the Calendar

Chapter 7

Euler's Generalization of Fermat's Theorem

7.1 Leonhard Euler

7.2 Euler's Phi-Function

7.3 Euler's Theorem

7.4 Some Properties of the Phi-Function

Chapter 8

Primitive Roots and Indices

8.1 The Order of an Integer Modulo n

8.2 Primitive Roots for Primes

8.3 Composite Numbers Having Primitive Roots

8.4 The Theory of Indices

Chapter 9

The Quadratic Reciprocity Law

9.1 Euler's Criterion

9.2 The Legendre Symbol and Its Properties

9.3 Quadratic Reciprocity

9.4 Quadratic Congruences with Composite Moduli

Chapter 10

Introduction to Cryptography

10.1 From Caesar Cipher to Public Key Cryptography

10.2 The Knapsack Cryptosystem

10.3 An Application of Primitive Roots to Cryptography

Chapter 11

Numbers of Special Form

11.1 Marin Mersenne

11.2 Perfect Numbers

11.3 Mersenne Primes and Amicable Numbers

11.4 Fermat Numbers

Chapter 12

Certain Nonlinear Diophantine Equations

12.1 The Equation $x^2 + y^2 = z^2$

12.2 Fermat's Last Theorem

Chapter 13

Representation of Integers as Sums of Squares

13.1 Joseph Louis Lagrange

13.2 Sums of Two Squares

13.3 Sums of More Than Two Squares

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Continued Fractions

15.1 Srinivasa Ramanujan

15.2 Finite Continued Fractions

15.3 Infinite Continued Fractions

15.4 Farey Fractions

15.5 Pell's Equation

Chapter 16

Some Modern Developments

16.1 Hardy, Dickson, and Erdős

16.2 Primality Testing and Factorization

16.3 An Application to Factoring: Remote Coin Flipping

16.4 The Prime Number Theorem and Zeta Function

Bibliography

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